IE406: Machine Learning Lab Assignment-7

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Lab-7: Eigenfaces for Face Recognition using PCA

The objective of this lab is to understand and implement **Principal Component Analysis (PCA)** for feature extraction from face images and apply it for a face recognition task using **Eigenfaces**. The performance of two classifiers, **K-Nearest Neighbors (KNN)** and **Diagonal Quadratic Discriminant Analysis (QDA)**, is evaluated.

DataSet Used :
AT&T ORL Face Dataset

- Contains 40 subjects with 10 images each
- Images have controlled lighting and pose

YALE Face Dataset

- Contains images with variations in lighting, expression, and alignment
- More challenging due to high intra-class variation

- The ORL dataset achieved significantly higher accuracy (up to 98.75% with KNN) compared to the YALE dataset, indicating that controlled conditions improve classification performance.
- The YALE dataset showed lower accuracy (around 78.79% with QDA), primarily due to variations in lighting and facial expressions.
- Increasing PCA dimensions from 20 to 100 slightly improved performance for ORL, but had minimal impact on YALE, indicating diminishing returns beyond a certain point.
- KNN performed consistently well on the ORL dataset, with k=3 giving the best results, suggesting optimal neighborhood size improves classification.

- QDA showed stable performance across different PCA dimensions but was generally lower than KNN, likely due to its Gaussian assumptions and sensitivity to variance estimation.
- For the YALE dataset, QDA slightly outperformed KNN, indicating that probabilistic modeling handled variation better in this case.
- The results demonstrate that PCA effectively reduces dimensionality while preserving important facial features for classification.
- Overall, dataset characteristics (lighting, alignment, expression) had a stronger impact on performance than the choice of PCA dimension.

Diagonal QDA (Manual Implementation)

- Assumes features are independent (diagonal covariance matrix)
- For each class:
    - Mean vector was computed
    - Variance of each feature was computed
    - Prior probability was calculated
- Prediction based on **maximum log-posterior probability**

Effect of PCA Dimensions

- Lower dimensions (d=20) result in loss of important features → lower accuracy
- Moderate dimensions (d=50, 100) improve performance
- Increasing dimensions beyond a point gives diminishing returns

ORL vs YALE Dataset

- ORL dataset achieved higher accuracy due to controlled conditions
- YALE dataset showed lower accuracy due to lighting and expression variations

KNN vs QDA

- KNN generally performed better due to its non-parametric nature
- QDA performance was affected by:
    - Gaussian assumption
    - Sensitivity to variance estimation